2.2.8 Aluminium chloride:

Preparation:

When aluminium metal or aluminium hydroxide is treated with hydrochloric acid, aluminium trichloride is formed. The reaction mixture is evaporated to obtain hydrated aluminium chloride.

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

$$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$$

McAfee Process:

Aluminium chloride is obtained by heating a mixture of alumina and coke in a current of chlorine.

$$Al_2O_3 + 3C + 3Cl_2 \longrightarrow 2AlCl_3 + 3CO_2$$

On industrial scale it is prepared by chlorinating aluminium around 1000 K

$$2Al + 3Cl_2 \xrightarrow{1000\ K} 2AlCl_3$$

Properties:

Anhydrous aluminium chloride is a colourless, hygroscopic substance.

An aqueous solution of aluminium chloride is acidic in nature. It also produces hydrogen chloride fumes in moist air.

$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

With ammonium hydroxide it forms aluminium hydroxide.

$$AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$$

With excess of sodium hydroxide it produces metal aluminate

$$AlCl_3 + 4NaOH \longrightarrow NaAlO_2 + 2H_2O + 3NaCl$$

It behaves like a Lewis acid and forms addition compounds with ammonia, phosphine and carbonylchloride etc... Eg. $AlCl_3.6NH_3$.

Uses of aluminium chloride:

1. Anhydrous aluminium chloride is used as a catalyst in Friedels Crafts reactions
2. It is used for the manufacture of petrol by cracking the mineral oils.
3. It is used as a catalyst in the manufacture on dyes, drugs and perfumes.

2.2.9 Alums:

The name alum is given to the double salt of potassium aluminium sulphate [$K_2SO_4.Al_2(SO_4)_3.24.H_2O$]. Now a days it is used for all the double salts with $M'_2SO_4.M''_2(SO_4)_3.24H_2O$, where M' is univalent metal ion or $[NH_4]^+$ and M'' is trivalent metal ion.

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